

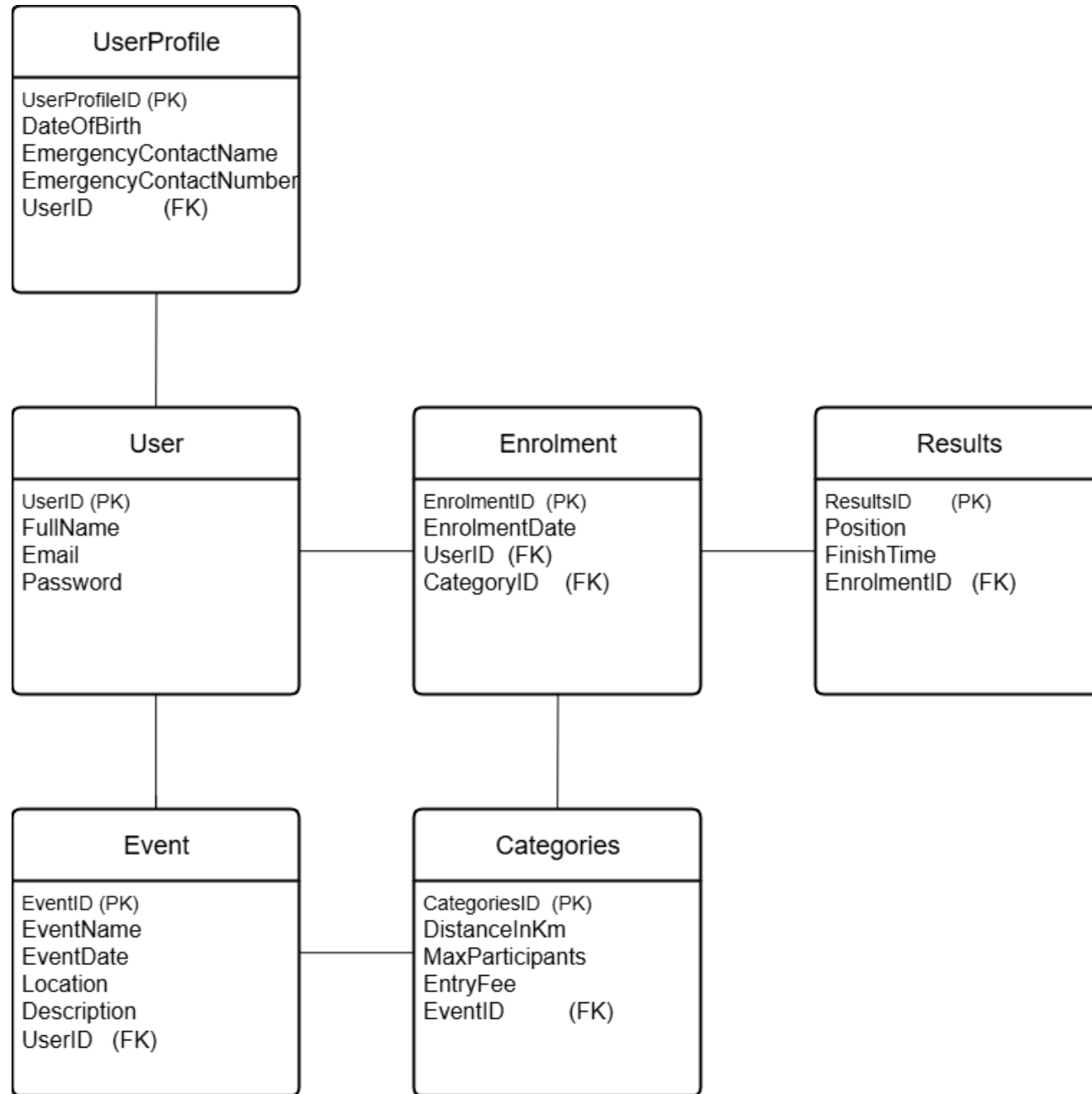


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Section A



Section B - RaceDay API Endpoint Plan

#	Method	Route	Description	Role Required	Request Body	Expected Response
1	POST	/api/auth/register	Creates a new User account as either an Organiser or a Participant.	None (public)	{ fullName, email, password, role }	201 Created – user object (no password); 400 Bad Request – validation failed
2	POST	/api/auth/login	Authenticates a User and returns a JWT access token.	None (public)	{ email, password }	200 OK – { token, user }; 401 Unauthorized – invalid credentials
3	GET	/api/users/me	Returns the logged-in User's account details and their UserProfile (if one exists).	Any (logged in)	None	200 OK – User + UserProfile object; 401 Unauthorized
4	POST	/api/users/me/profile	Creates the UserProfile for the logged-in User (one per User – enforced by the UNIQUE UserID).	Any (logged in)	{ dateOfBirth, emergencyContactName, emergencyContactNumber }	201 Created – UserProfile object; 409 Conflict – profile already exists
5	PUT	/api/users/me/profile	Updates the logged-in User's existing UserProfile.	Any (logged in)	{ dateOfBirth, emergencyContactName, emergencyContactNumber }	200 OK – updated UserProfile; 404 Not Found – no profile yet
6	GET	/api/users/{id}	Returns public details for a specific User (e.g. an organiser's name on an Event page).	Any (logged in)	None	200 OK – User object (FullName only); 404 Not Found
7	GET	/api/events	Lists all Events, with optional filters (EventDate, Location).	None (public)	None	200 OK – array of Event objects
8	GET	/api/events/{id}	Returns full details for one Event, including its Categories.	None (public)	None	200 OK – Event object with nested Categories; 404 Not Found
9	POST	/api/events	Creates a new Event. UserID is taken from the logged-in Organiser's token.	Organiser	{ eventName, eventDate, location, description }	201 Created – Event object; 400 Bad Request
10	PUT	/api/events/{id}	Updates an Event's details. Only the User (UserID) who owns it may edit it.	Organiser (owner)	{ eventName, eventDate, location, description }	200 OK – updated Event; 403 Forbidden; 404 Not Found
11	DELETE	/api/events/{id}	Deletes an Event. Its Categories must be deleted first (no cascade).	Organiser (owner)	None	204 No Content; 403 Forbidden; 409 Conflict – Categories still exist
12	GET	/api/events/{eventId}/categories	Lists all Categories belonging to a specific Event.	None (public)	None	200 OK – array of Categories
13	GET	/api/categories/{id}	Returns details for one Category.	None (public)	None	200 OK – Category object; 404 Not Found
14	POST	/api/events/{eventId}/categories	Adds a new Category to an Event.	Organiser (owner of Event)	{ distanceInKm, maxParticipants, entryFee }	201 Created – Category object; 403 Forbidden; 404 Not Found
15	PUT	/api/categories/{id}	Updates a Category's details.	Organiser (owner of parent Event)	{ distanceInKm, maxParticipants, entryFee }	200 OK – updated Category; 403 Forbidden; 404 Not Found
16	DELETE	/api/categories/{id}	Removes a Category from an Event.	Organiser (owner of parent Event)	None	204 No Content; 403 Forbidden; 409 Conflict – Enrolments still exist
17	POST	/api/categories/{categoryId}/enrolments	Enrols the logged-in User into a Category. UserID comes from the token, CategoryID from the route.	Participant	{ } (EnrolmentDate defaults automatically)	201 Created – Enrolment object; 404 Not Found; 409 Conflict – already enrolled
18	GET	/api/users/me/enrolments	Lists all of the logged-in Participant's Enrolments.	Participant	None	200 OK – array of Enrolments
19	GET	/api/categories/{categoryId}/enrolments	Lists every User enrolled in a Category.	Organiser (owner of parent Event)	None	200 OK – array of Enrolments; 403 Forbidden

#	Method	Route	Description	Role Required	Request Body	Expected Response
20	DELETE	/api/enrolments/{id}	Cancels/removes an Enrolment. Its Result must be deleted first (no cascade).	Participant (owner)	None	204 No Content; 403 Forbidden; 404 Not Found; 409 Conflict – Result still exists
21	POST	/api/enrolments/{enrolmentId}/results	Captures the Result for an Enrolment (one per Enrolment – enforced by the UNIQUE EnrolmentID).	Organiser (owner of parent Event)	{ position, finishTime }	201 Created – Results object; 403 Forbidden; 404 Not Found; 409 Conflict – Result already recorded
22	GET	/api/categories/{categoryId}/results	Returns the leaderboard for a Category, ordered by Position.	None (public)	None	200 OK – array of Results joined with User FullName; 404 Not Found
23	GET	/api/users/me/results	Returns the logged-in Participant's own race history.	Participant	None	200 OK – array of Results
24	PUT	/api/results/{id}	Corrects a previously recorded Result.	Organiser (owner of parent Event)	{ position, finishTime }	200 OK – updated Results object; 403 Forbidden; 404 Not Found

Section C

```
CREATE DATABASE RaceDayDB;

USE RaceDayDB;

CREATE TABLE USERS(
    USERID INT IDENTITY (1,1) PRIMARY KEY,
    FULLNAME VARCHAR(100) NOT NULL,
    EMAIL VARCHAR (150) NOT NULL,
    PASSWORD VARCHAR(255) NOT NULL,
    ROLE VARCHAR (20) NOT NULL
);

ALTER TABLE USERS
ADD CONSTRAINT UQ_Users_Email UNIQUE (EMAIL);

INSERT INTO USERS(FULLNAME, EMAIL, PASSWORD, ROLE)
VALUES ('Sarah Nkosi', 'sarah.nkosi@raceday.co.za', 'hashed_pw_1', 'Organiser'), -- UserID 1
       ('Mike van Wyk', 'mike.vanwyk@raceday.co.za', 'hashed_pw_2', 'Organiser'), -- UserID 2
       ('Thabo Mokoena', 'thabo.mokoena@example.com', 'hashed_pw_3', 'Participant'), -- UserID 3
       ('Emma Botha', 'emma.botha@example.com', 'hashed_pw_4', 'Participant'); -- UserID 4

SELECT * FROM USERS

CREATE TABLE USERSPROFILE (
    USERPROFILEID INT IDENTITY(1,1) PRIMARY KEY,
    DATEOFBIRTH DATE NULL,
    EMERGENCYCONTACTNAME VARCHAR(100) NULL,
    EMERGENCYCONTACTNUMBER VARCHAR(20) NULL,
    USERID INT NOT NULL,
    CONSTRAINT FK_USERPROFILE_USER FOREIGN KEY (USERID)
    REFERENCES [USERS](USERID)
);

INSERT INTO USERSPROFILE(USERID, DATEOFBIRTH, EMERGENCYCONTACTNAME, EMERGENCYCONTACTNUMBER)
VALUES (3, '1995-03-14', 'Lindiwe Mokoena', '0827654321'),
       (4, '1998-07-22', 'Johan Botha', '0839876543');

SELECT * FROM USERSPROFILE
```

```

CREATE TABLE EVENTS (
EVENTID INT IDENTITY(1,1) PRIMARY KEY,
EVENTNAME VARCHAR(150) NOT NULL,
EVENTDATE DATE NOT NULL,
LOCATION VARCHAR(150) NOT NULL,
DESCRIPTION VARCHAR(500) NOT NULL,
USERID INT NOT NULL,
CONSTRAINT FK_EVENT_USER FOREIGN KEY (USERID)
REFERENCES [USERS](USERID)
);

```

```

INSERT INTO EVENTS(EVENTNAME, EVENTDATE, LOCATION, DESCRIPTION, USERID)
VALUES ('Johannesburg City Marathon', '2026-10-18', 'Johannesburg, Gauteng', 'Annual road race through the city centre.', 1),
('Sandton Fun Run', '2026-11-02', 'Sandton, Gauteng', 'Family-friendly fun run and walk.', 1),
('Cape Town Trail Challenge', '2026-09-27', 'Cape Town, Western Cape', 'Off-road trail running on Table Mountain.',

```

```

SELECT * FROM EVENTS

```

```

CREATE TABLE CATEGORY (
CATEGORIESID INT IDENTITY(1,1) PRIMARY KEY,
DISTANCEINKM DECIMAL(5,2) NOT NULL,
MAXPARTICIPANTS INT NOT NULL DEFAULT 100,
ENTRYFEE DECIMAL(8,2) NOT NULL DEFAULT 0,
EVENTID INT NOT NULL,
CONSTRAINT FK_CATEGORIES_EVENT FOREIGN KEY (EVENTID)
REFERENCES EVENTS(EVENTID)
);

```

```

INSERT INTO CATEGORY(DISTANCEINKM, MAXPARTICIPANTS, ENTRYFEE, EVENTID)
VALUES (10.00, 500, 150.00, 1), -- 10km, Marathon JOZI
(21.10, 300, 250.00, 1), -- 21km, Marathon JOZI
(42.20, 200, 350.00, 1), -- 42km, Marathon JOZI
(5.00, 400, 80.00, 2), -- 5km, Sandton Fun Run
(2.00, 200, 40.00, 2), -- 2km, Sandton Fun Run
(15.00, 150, 200.00, 3), -- 15km, CT Trail
(30.00, 100, 300.00, 3); -- 30km, CT Trail

```

```

SELECT * FROM CATEGORY

```

```
CREATE TABLE ENROLMENTS (  
  ENROLMENTID INT IDENTITY(1,1) PRIMARY KEY,  
  ENROLMENTDATE DATETIME NOT NULL DEFAULT GETDATE(),  
  USERID INT NOT NULL,  
  CATEGORYID INT NOT NULL,  
  CONSTRAINT FK_ENROLMENT_USER FOREIGN KEY (USERID)  
  REFERENCES [USERS](USERID),  
  CONSTRAINT FK_ENROLMENT_CATEGORIES FOREIGN KEY (CATEGORYID)  
  REFERENCES Category(CATEGORIESID),  
  CONSTRAINT UQ_ENROLMENT_USER_CATEGORY UNIQUE (USERID, CATEGORYID)  
);
```

```
INSERT INTO ENROLMENTS (USERID, CATEGORYID)  
VALUES (3, 1), -- Thabo -> 10km Marathon JOZI  
       (3, 6), -- Thabo -> 15km CT Trail  
       (4, 2), -- Emma -> 21km Marathon JOZI  
       (4, 4); -- Emma -> 5km Sandton Fun Run
```

```
SELECT * FROM ENROLMENTS
```

```
CREATE TABLE RESULT (  
  RESULTSID INT IDENTITY(1,1) PRIMARY KEY,  
  POSITION INT NULL,  
  FINISHTIME TIME NULL,  
  ENROLMENTID INT NOT NULL UNIQUE,  
  CONSTRAINT FK_RESULTS_ENROLMENT FOREIGN KEY (ENROLMENTID)  
  REFERENCES ENROLMENTS(ENROLMENTID)  
);
```

```
INSERT INTO RESULT(ENROLMENTID, POSITION, FINISHTIME)  
VALUES (1, 12, '00:52:31'),  
       (3, 5, '01:58:47');
```

```
SELECT * FROM RESULT
```